



05 — Consolidation (Prototypes → Schemas)

Overview

Consolidation lifts replayed latent prototypes into first-class symbolic schemas stored in SQLite with FTS indexing. Two paths exist: the **brain-only latent path** (zero LLM calls) and a legacy text path. The latent path is the production default.

Latent Consolidation Pipeline

Step 1: Prototype Selection

Prototypes are selected if they have accumulated enough support (member episodes) and haven't been consolidated recently.

Step 2: Latent Schema Building

For each selected prototype p with centroid $\mathbf{c}$ and member episodes $\{e_1, \dots, e_m\}$:

Content text is derived from the prototype's lexical signature and member episode texts — no LLM involved.

Embedding: the prototype centroid $\mathbf{c}$ becomes the schema's embedding.

Facets extracted from episode metadata:

- `memory_layer` : inferred from schema type (profile / domain / workspace)
- `source_kind` : "consolidation" or "explicit_remember"
- `entity_count` , `episode_count` : from prototype support

Step 3: Schema Classification

Consolidated schemas are classified by structure:

```
if source_kind == "explicit_remember" → no reclassification
elif sentence_count >= 3 or len(text) > 300 → "episodic_summary"
else → "fact"
```

Step 4: Supersession & Contradiction Check

For each new schema claim, check against existing active schemas:

Cosine similarity check:

$$\text{cos}_{\text{new,old}} = \langle \mathbf{e}_{\text{new}}, \mathbf{e}_{\text{old}} \rangle$$

Thresholds from `supersession_manifold.py`:

Condition	Action
Same scope, $\text{cos} \geq 0.85$	Flag for review (potential supersession)
Same scope, $\text{cos} \in [0.70, 0.85)$	Check direction score
Same scope, direction $\geq$ <code>DIRECTION_THRESHOLD</code>	Supersede old schema
Cross-scope, $\text{cos} \geq 0.78$	Reinforce + record cross-scope evidence
$\text{cos} \geq 0.35$	Mark as topically related

Direction score (see §10 — Supersession Manifold):

$$\text{dir} = \langle \hat{\mathbf{d}}, \mathbf{v}_{\text{SVD1}} \rangle, \quad \mathbf{d} = \mathbf{e}_{\text{new}} - \mathbf{e}_{\text{old}}$$

Step 5: Schema Storage

New schemas are inserted into the `schemas` table with:

- `content_text` : canonicalized text
- `embedding` : float32 L2-normalized vector
- `facets_json` : JSON metadata dictionary
- `status` : "active", "needs_review", or "superseded"

- `created_at` , `last_updated_at` : timestamps
- FTS5 index updated automatically via triggers

Step 6: Schema-Prototype Mapping

A row is inserted into `schema_prototype_map` linking the new schema to its source prototype, enabling bidirectional navigation between latent and symbolic layers.

Schema Decay

Active schemas that have never been recalled can be decayed:

$$s_{\text{schema}} \leftarrow s_{\text{schema}} \cdot \gamma_{\text{decay}}, \quad \gamma_{\text{decay}} \in (0, 1]$$

Triggered by `decay_schemas(idle_days=30.0)` — the schema store's `decay_unused` method applies a salience penalty to schemas with zero recall count and `last_recalled_at` older than `idle_days`.

Configuration

Consolidator Constructor

Parameter	Type	Default	Description
<code>max_episodes_per_prototype</code>	<code>int</code>	<code>8</code>	Max member episodes used for text extraction

Supersession Thresholds (Global Constants)

Constant	Value	Description
<code>SAME_SCOPE_COS_THRESHOLD</code>	<code>0.85</code>	Cosine threshold for same-scope action
<code>EXTENDED_SAME_SCOPE_COS_THRESHOLD</code>	<code>0.70</code>	Lower bound for same-scope direction check

Constant	Value	Description
CROSS_SCOPE_COS_THRESHOLD	0.78	Cosine threshold for cross-scope linking
DIRECTION_THRESHOLD	0.10	Minimum direction score for supersession
DIR_REVIEW_BAND	(-0.05, 0.05)	Direction range triggering needs_review
TOPICAL_THRESHOLD	0.35	Minimum similarity for topical relationship

Key Invariants

- 1. Zero LLM calls in the consolidation path — schemas are built from prototype geometry and lexical signatures.
- 2. Explicit memories (source_kind = "explicit_remember") are never reclassified.
- 3. Cross-scope schemas are never superseded — only reinforced with cross-scope evidence.
- 4. Schemas with status = "superseded" are excluded from context/recall by default.